KLOUDSPOT | TECHNICAL ASSESSMENT

Project 1: Employee Management System

Position	Software Engineer / Backend (Freshers)
Assessment duration	24 hours from receipt of assessment
Technical point of contact	Manjunatha | manjunatha.v@kloudspot.com
Mandatory HR recipient	hr.team@kloudspot.com

1. Objective

Build a web application that allows an administrator to manage employee records through a simple and intuitive user interface. The application must support complete CRUD operations using a frontend, REST APIs, and a database.

2. Functional requirements

A. Employee management

Each employee record must contain: Employee ID, Full Name, Email Address, Phone Number, Department, Designation, Date of Joining, Salary, and Employment Status (Active or Inactive).

- Add a new employee; view employee details; edit employee information.
- Delete an employee with confirmation; list all employees.
- Search employees by name or email.
- Filter by department and employment status.
- Implement pagination for the employee list.

B. User interface

- Responsive Employee List page.
- Add Employee page/form.
- Edit Employee page/form.
- Employee Details page.

C. Backend

Implement REST APIs to support the required employee management operations and integrate them with the frontend and persistent storage.

D. Database

Use MongoDB, MySQL, or PostgreSQL.

3. Technology stack

- Backend: Java 17 or later, Spring Boot, Spring Data JPA or Spring Data MongoDB, Maven.
- Frontend: Any frontend technology of your choice.
- Testing and documentation: JUnit 5, Mockito, Swagger/OpenAPI, Git.

4. Deliverables

- Complete backend source code and frontend source code.
- Database configuration and required setup scripts.

- Unit tests covering service-layer business logic.
- README with prerequisites and setup/run instructions.
- Sample API requests or a Postman collection.
- Screenshots or a short demonstration of the working application.

5. Optional enhancements (bonus points)

- Login and basic authentication.
- Department management.
- Export employee data to CSV.
- Docker Compose setup.
- Automated frontend tests.

6. Submission instructions

- Complete and submit the assessment within 24 hours of receiving it.
- Reply to the assessment email with your GitHub repository link or attach a ZIP file containing the source code and required deliverables.
- Address your submission to manjunatha.v@kloudspot.com and mandatorily include hr.team@kloudspot.com in the recipients (To or CC).
- Ensure the application runs locally by following the README without requiring manual code changes.

Please ensure the implementation is complete, properly organized, and accompanied by the required documentation.